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Numerical simulation of the behavior of lithium-ion battery electrodes during the calendaring process via the discrete element method

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ABSTRACT

Calendering is a key process step in the production chain of lithium ion battery electrodes since it strongly affects the microstructure and micromechanics of the electrodes and hence, the performance and life of the battery. A comprehensive understanding is therefore necessary to find optimal levels of calendering which can help to enhance conductive and mechanical properties. Within this context, this study proposes a novel discrete element method (DEM) approach which can capture the mechanical properties of single Li $[Ni_{1/3}Mn_{1/3}Co_{1/3}]O_2$ (NMC) particles with an appropriate elasto-plastic contact model, as well as the mechanical behavior of the additive-binder matrix via an additional bond model [1]. With the support of real produced cathodes, the simulations were able to reproduce the calendering process while providing detailed information about the changes in electrode structural and mechanical parameters. In particular, the investigated features comprise the electrode porosity and thickness along with the specific free surface area, the contact area between NMC particles and current collector, the coordination number of NMC particles, the number of broken bonds and the directionality of the contacts together with the generated stress within the electrode. Moreover, the simulations are able to capture the viscoelastic response of the electrode, showing that the relative elastic recovery can be almost up to 17%, an important piece of information that cannot be obtained experimentally to date. Having established the fundamentals and simulation feedback, an upcoming publication is meant to complete this research by providing a numerical overview of the relations between the electrode structure and its properties affected by the calendering process and during the first electrochemical cycles.

1. Introduction

Lithium ion batteries have long been considered as an electrochemical power source for a wide range of applications due to their long life cycles and high-rate capabilities. Regarding the production process, the calendering step has been proved to play a fundamental role since it induces important changes in the microstructure of the electrode and hence, influences the performance and life of the battery [2–4]. Calendering consists of compacting the electrodes and is performed after electrode coating and drying. As electrodes are compressed, the number of particle-to-particle and particle-to-current collector contacts increases, mainly leading to an improvement in electrical and thermal conductivity [5]. Furthermore, this production process also contributes to a reduction in electrode thickness at the same active material loading, which translates into an increase of volumetric energy density. Nevertheless, the decrease in electrode porosity lowers the inner specific area of the electrode and enlarges the electrode tortuosity [6]. Additionally, high compression loads can cause high mechanical levels of stress on active material particles, which is ultimately unfavorable to cell performance and cell life [7]. Therefore, it is crucial for electrode production to find optimal levels of calendering in order to compensate conductive and mechanical properties.

To date, several studies have been dedicated to this production step. Zheng et al. [3] investigated the calendering effects on the physical and electrochemical properties of $Li[Ni_{1/3}Mn_{1/3}Co_{1/3}]O_2$ (NMC) cathodes, showing that calendering implies an increase in the first coulombic efficiency but a decrease in specific capacity. Meyer et al. [8] confirmed an exponential relation between the applied line load and the achieved coating density and porosity for both positive and negative electrodes. Moreover, they were able to identify that the compaction resistance of the electrode coating mainly depends on the chosen material and formulation rather than on the roller speed. However, in order to establish all specific correlations, a large number of experiments would need to be performed, which would comprise an intensive cost- and time-procedure. Taking this into consideration, numerical approaches have recently been acclaimed since they constitute a useful tool not only to provide additional information, but also to substitute future experimental work. In fact, several studies have already been performed with the

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aim of modelling and numerically simulating lithium-ion electrodes during calendering. By proposing a Finite Element Method (FEM) approach, Wang et al. [9] implemented a model and analyzed the effects of friction and deformability of graphite particles on the compressibility of lithium-ion anodes. Kespe et al. [10] developed a spatially three-dimensional model of a lithium-ion battery half-cell on the microscale and used it to simulate spherical active material particles under compression. Also in this line of work, Ott et al. [7] investigated the influence of calendering on the percolation threshold and the effective conductivity. They proposed a discrete element modelling strategy to densify the microstructure using the Hertz contact model and applied a resistor-network model to subsequently determine the effective conductivity. A pseudo 2-dimensional (P2D) model was introduced by Lenze et al. [11] to study the calendering impacts on battery cell performance by combining experimental methods with physico-chemical simulation. Even though these previous works have helped to give insight into the calendering process, there are still certain structural and mechanical aspects that remain unknown or not covered by simulations. For instance, it would be of great interest to be able to estimate the elastic recovery of the electrode coating after it passes through the calendering rolls because it would help to predict the calendering degree needed for specific final structural characteristics. To date, this feature cannot be assessed experimentally due to the extremely small distance between the rolls and the remarkably high resolution required to measure the actual position of the rolls.

Within this framework, the goal of this study is to propose a simulation method to analyze the calendering process directly from a microscale level. The macroscale response of the electrode is derived by capturing the behavior of single active material particles and their interactions within the additive-binder matrix. In order to achieve this goal, the discrete element method (DEM) is taken as a well-founded tool to simulate such particulate composites - a method which has already been proved to be advantageous in previous studies [1,7,12,13]. The presented simulations are combined with experiments to calibrate and validate the proposed method. Firstly, the mechanical behavior of single NMC active material particles is investigated via nanoindentation. By using this experimental data, an appropriate contact model is defined and implemented together with a bond model [1] for the simulations on the electrode level. Moreover, continuously coated NMC cathodes are produced and subjected to four different calendering degrees using a continuous calender. By means of these real calendering experiments, the simulations are calibrated and validated, which brings the possibility to characterize additional features and get deeper knowledge into the calendering process. In fact, this line of work is followed in an upcoming publication. Having defined the fundamentals and the possible outcomes of a simulation, this second publication focuses on the definition of the so-called key values. The key values comprise several structural and mechanical features at particle level and are related to important electrode-scale properties such as the electrical or ionic conductivity, which at the same time dominate overall cell properties. The ultimate goal is therefore to establish a relation between the structural, the mechanical and the transport properties in view of better understanding the overall electrode response during the calendering process.

2. Material and experimental methods

For the validation of the simulations, a NMC-based cathode was produced and subjected to different calendering line loads. Conductive graphite and carbon black were used as conducting additives and polyvinylidene fluoride (PVDF) as binder. The NMC powder (90%-wt.) was dry mixed in a planetary mixer (PMH-10, Netzsch Cop.) with the conductive graphite (2%-wt.), carbon black (4%-wt.) and PVDF (4%-wt.) powder for 5 min. *N*-methyl-2-pyrrolidone (NMP) was used as solvent and added for the subsequent wet mixing process. The suspension, with a solids content of 45%-wt., was dispersed for 60 min and degassed under vacuum for another 30 min using a planetary mixer. The following coating on the aluminum substrate and subsequent drying process was performed using a continuous indirect comma-bar coating device and convection dryer (LabCo, Kroenert Corp). The electrode was coated double-sided to take advantage of the symmetry of the system in the simulation. The selected temperatures of the three-stage drying profile were 80 °C, 100 °C and 120 °C. Within this procedure, the coating and drying speed were kept constant at 2 m min^{-1} as reported in [14]. The final mass loading of the coating layer was 16.88 ± 0.12 mg cm^{-2}. For the calendering step, a pilot scale two roll compactor GKL 400 (Saueressig GmbH Co. KG) was used with rolls of 430 mm in diameter and 465 mm in width. Additional details regarding the equipment and the analysis of the calendering force can be found in [8]. For this study, four different line loads were applied by varying the gap size between the rolls. The velocity of the rolls was kept constant at 2 m min^{-1}. The electrode thickness was measured before and directly after the calendering step by using a digital gauge (ID-C, Mitutoyo). Likewise, the porosity of the non-calendered and calendered samples was calculated from the mass loading, the measured thickness and the mean physical density of the components, as it is explained in [6]. The properties are gathered in Table 1 for the non-calendered and the resulting four calendered cathodes. The thickness includes both sides of the electrode coating and the current collector. The thickness of the aluminum substrate was approximately 23.5 μm.

The importance of reproducing the real mechanical behavior of a NMC particle is highlighted in Section 6 in terms of choosing the appropriate contact model for the DEM simulation. Bearing this in mind, nanoindentation experiments were performed on single active material particles (TriboIndenter TI 900 from Hysitron, Inc. Minneapolis). The compressions were carried out with a constant loading and unloading rate of 30 nm s^{-1}. A total of 40 particles were analyzed and a force-displacement curve (F-d) was obtained during each measurement. Additionally, the particle size distribution of NMC was analyzed by means of laser diffraction (Helos Sympatec, Germany). Scanning electron microscopy pictures (LEO Gemini 1550 Carl Zeiss, Germany) were performed to support decisions regarding the simulation fundamentals.

3. Results on the microscale level: single NMC particles

As cited, this study proposes a simulation method to analyze the calendering process directly from a microscale level. For this reason, the interaction between NMC particles needs to be well understood in the first place in order to derive the macroscale response of the electrode. The Hertz contact model [15] is the most common model for DEM simulations because it considers a relatively low number of input parameters, does not need an outstanding calculation time and delivers valid results for non-linear elastic materials. Nevertheless, it was determined via nanoindentation experiments and supported with scanning electrode microscopy pictures, that $Li[Ni_{1/3}\ Mn_{1/3}\ Co_{1/3}]O_2$ (NMC) active material particles are elasto-plastic in nature. They behave elastically (Fig. 1c) under very low levels of compaction (up to a displacement approximately 0.10% of their radius) and the force-displacement curve (F-d) can be well fitted by using the theory of Hertz. In the case of higher

Table 1
Relevant structural and process parameters of the produced non-calendered and calendered electrodes.

Parameter	Symbol	Non-calendered	C1	C2	C3	C4
Thickness [μm]	h	180	151.2	142.2	129.2	118.6
Porosity [−]	ε	0.522	0.416	0.372	0.295	0.217
Coating density [g cm^{-3}]	ρ	2.13	2.61	2.8	3.15	3.49
Normal calendering stress [MPa]	P_{Cal}	–	19.82	42.71	79.13	159.37
Gap between rolls [μm]	h_{Rolls}	–	130	108	80	20

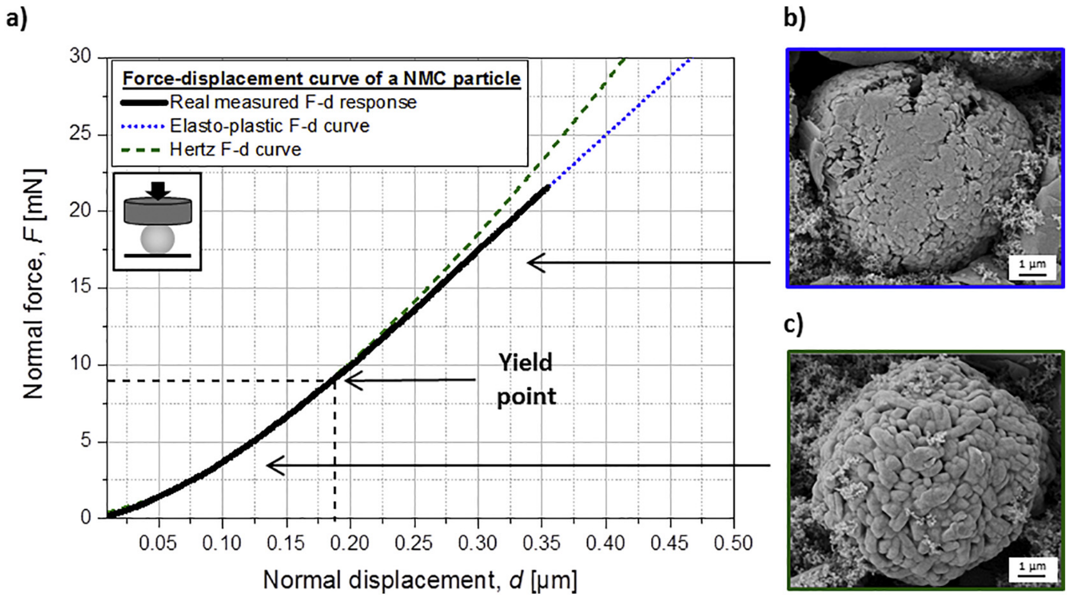

Fig. 1. (a) Exemplary force displacement curve (F-d) of a NMC particle during a nanoindentation experiment; (b) SEM image of a NMC particle subjected to a low compression force under the yield point; (c) SEM image of a NMC particle subjected to a high compression force beyond the yield point.

compaction forces beyond the yield point (Fig. 1b), there is a plastic contribution that cannot be disregarded as the experimental F-d curve deviates increasingly from the theoretical Hertz curve (Fig. 1a). Therefore, the contact model of Hertz overestimates their mechanical behavior. As a result, the strategy adopted in this work was to include the effect of plasticity based on the theory of elasto-plastic particle contacts from Thornton and Ning [16]. This approach firstly calculates the elastic contact deformation according to Hertz [15] and considers an elasto-plastic deformation when the yield point is reached. The unloading is performed subsequently according to Hertz theory.

The described behavior can be transferred for the case of particle-particle contact, considering two particles of radii r_i, elastic moduli E_i and Poisson ratios ν_i $(i=1{,}2)$. At the beginning of the contact, the elastic force is calculated as [15]:

$$F_{el} = \frac{4}{3} E^* \sqrt{R^* \delta^3} \tag{1}$$

where δ is the overlap, E^* is the effective elastic modulus obtained from E_i and ν_i, and R^* is the effective radius defined from r_i, as explained in [1]. The contact becomes plastic if the contact radius a exceeds the contact radius at which the yield point occurs, a_y. The contact radius can be obtained from the acting force as:

$$a = \left(\frac{3F_{el}R^*}{4E^*}\right)^{1/3} \tag{2}$$

The value of a_y is determined by the critical yield pressure p_y, which represents the maximum contact pressure for the onset plastic deformation. p_y can be related to the critical yield normal force F_y and the critical yield normal contact displacement $\delta_y$through [17]:

$$F_y = \frac{1}{6}\left(\frac{R^*}{E^*}\right)^2 \left(\pi p_y\right)^3 \tag{3}$$

$$\delta_y = \frac{1}{4}\frac{R^*}{(E^*)^2}\left(\pi p_y\right)^2 \tag{4}$$

Once the yield point is reached, the plastic deformation is initiated and the contact pressure distribution changes from the elastic elliptical shape to a uniform pressure distribution as the force increases. In the elasto-plastic zone, the force-displacement relationship is expressed as [18]:

$$F_{el-pl} = F_y + \pi p_y R^* (\delta - \delta_y) \tag{5}$$

which indicates a linear relationship between the contact normal force and the contact displacement. The calculation of the unloading force depends on the history of the contact. If the yield point is not reached, the Hertz model is followed strictly. On the other hand, if the critical yield pressure is exceeded, the unloading force is calculated according to the plastic effective radius R_p^*, which is determined using the effective Hertzian force and the plastic force at the point of unloading as shown in [18]. The contact force-displacement relation can be therefore expressed in the form of

$$F_{unloading} = \frac{4}{3} E^* \sqrt{R_p^* (\delta - \delta_R)} \tag{6}$$

where δ_R represents the residual plastic displacement, as introduced in [18].

A total of 40 particles with different diameters were experimentally analyzed via nanoindentation and the yield point was determined for all of them in order to assess the influence of particle size. Fig. 2 shows the critical normal displacement (δ_y) that defines the offset of plastic deformation for the analyzed particles. It can be seen that δ_y is indeed dependent on the particle size, being a linear approximation reasonably valid with a coefficient of determination equal to 0.89. The established correlation is of great importance since NMC particles used for the production of the cathode do not have the same size. In fact, the laser diffraction measurements revealed a mean particle size of 9.35 μm and a $x_{10,3}$ and a $x_{90,3}$ of 4.96 and 16.59 μm, correspondingly (Table 2). The polydispersity of the system was also considered in the simulations, as it is detailed in Section 4. The slope of this correlation, named yield ratio (YR), was regarded in the simulation as the input parameter for the outset of the elasto-plastic model. Remaining input characteristics of NMC particles are listed in Table 2.

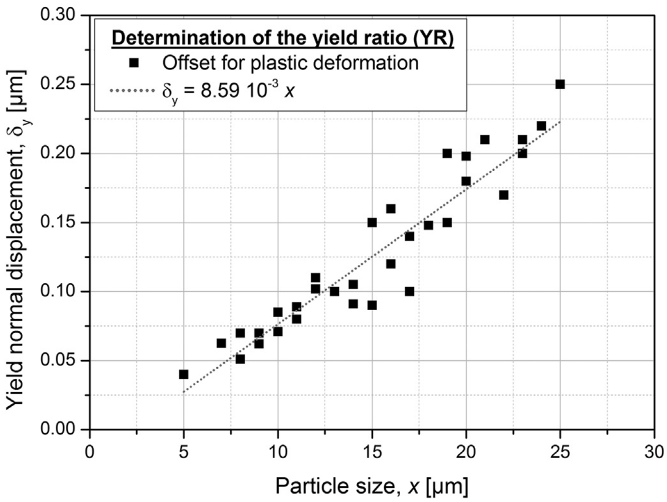

Fig. 2. Determination of the yield ratio (YR) on the basis of the correlation between the critical normal displacement and the particle size.

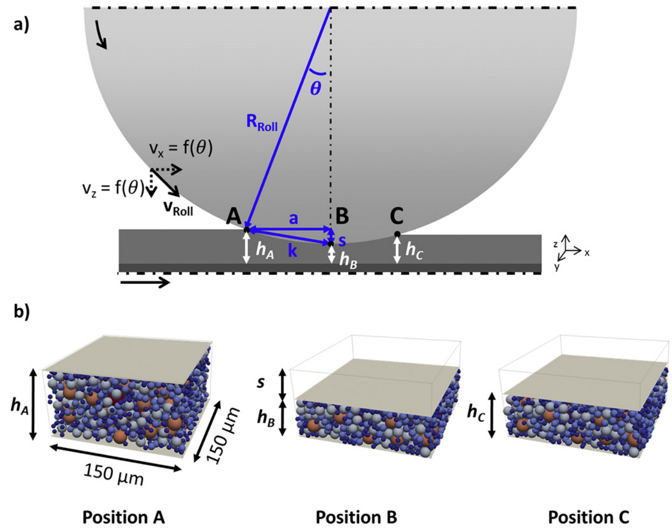

Fig. 3. a) Outline of the calendering process. The symmetry of the system is taken into consideration and only one side of the electrode is displayed. Likewise, half of the roll is presented. The electrode thickness is magnified to better discern the geometrical parameters; b) Simulation set-up and outline of positions A,B and C.

4. Simulation set-up of the calendering process of NMC-based electrodes

Physical and geometrical characteristics of the real calendering process were investigated previous to defining the simulation set-up. In order to clarify the statements, Fig. 3a illustrates a schematic of the calendering process. It must be underlined that this picture is meant to complement the text and the size relation does not correspond to the reality as it will be explained in brief. When discussing the electrode thickness, it is always referred to the thickness of one side of the coating. The first aspect examined was the angle of contact point between the roll and the electrode, θ, when they first become in touch (position A in Fig. 3a). For the geometrical calculation of θ, it was considered that the electrode thickness in between the rolls (h_B in Fig. 3a) is the same as the final thickness (h_c in Fig. 3a) even though the electrode exhibits a viscoelastic behavior in reality, i.e., h_C is greater than h_B [22,23]. This assumption had to be made because this difference in thickness cannot be experimentally measured as a rule. Nevertheless, this deviation would differ the calculations scantily and thus, this premise was considered to be reasonable. Considering $s = h_A - h_B$, θ can be estimated as:

$$\theta = \cos^{-1}\left(\frac{R_{Roll} - s}{R_{Roll}}\right) \tag{7}$$

Depending on the calendering degree, the value of s ranged from 14.4 μm for electrode C1 and 30.7 μm for electrode C4. θ was in all cases lower than 1°, being 0.66° for electrode C1 and 0.96° for electrode C4. This calculation proves that the contact between the electrode and the roll surface takes place mainly in normal direction as a consequence of the clear contrast between the radius of the roll (215 mm) and the electrode thickness. Taking this into account, it was decided to simulate a horizontal plate located above the particles, as presented in Fig. 3b. The second aspect to be considered is the velocity of the calendering process. As it has already been mentioned in Section 5, the velocity of the rolls was set constant and equal to 2 m min^{-1}. If the absolute velocity is decomposed in its directional components, it can be stated that the relative velocity in X direction between the electrode and the roll is practically zero. According to Fig. 3a, the velocity in Z direction can be calculated geometrically as a function of θ, being greater in position A and becoming zero in position B:

$$v_{z,A} = \frac{s}{k} v_{Roll} \tag{8}$$

$$v_{z,B} = 0 \tag{9}$$

The calculated $v_{z,A}$ was very low in all cases; 0.011 m min^{-1} for electrode C1 and 0.017 m min^{-1} for electrode C4. Consequently, the dynamic regime was ensured even if the velocity of the whole process was left constant and equaled to $v_{z,A}$ (variations in kinetic energy by simulating a velocity profile or a constant $v_{z,A}$ proved to be insignificant). Considering this outcome and in order to speed up the simulations, the plate moved downwards in the direction of the electrode with a velocity equivalent to the calender roll $v_{z,A}$. From position B to C, the velocity moving upwards had the same value but in opposite direction.

The electrode initial structure was generated using the open source DEM simulation software LIGGGHTS [24]. Based on the laser diffraction measurements, five representative particle sizes were chosen for the simulation, namely 5, 7.5, 10.5, 15 and 18 μm. The number of particles of each size was calculated regarding the density distribution results and the electrode initial composition, porosity and thickness. The properties of the particles corresponded to those of NMC (Table 2). Due to computational limitations, it would be too costly to simulate a complete electrode on the microscale. As a part of a preliminary sensitivity study, it became discernible that a representative volume element (RVE) of $150 \times 150 \times h_A$ μm^3 was sufficient to assure valid and representative results while maintaining reasonable computational time. The boundaries of the RVE in x and y direction were held periodic. Regarding the z dimension, the length corresponded to the thickness of the non-calendered electrode (see Fig. 3b – Position A). A plate with aluminum properties was placed on the bottom in order to represent the current collector. On the top, as already explained and shown in Fig. 3a

Table 2
Input parameters for the characterization of NMC particles.

Parameter	Value	Source
Young's modulus [GPa]	142	[19,20]
Poisson ratio [−]	0.25	[21]
Coefficient of restitution [−]	0.25	[21]
Density [g cm^{-3}]	4.75	[21]
Particle size [μm]		Measured
$x_{10,3}$	4.96	
$x_{50,3}$	9.35	
$x_{90,3}$	16.59	
Yield ratio [−]	$8.59 \cdot 10^{-3}$	Measured

horizontal plate was located with steel properties in order to describe the roll surface. The simulation started by randomly inserting a total of 2237 spherical NMC particles with the constraint that they were not in contact. The binder-additive matrix, composed of small additive particles (carbon black and graphite) embedded in PVDF acting as the binder (see Section 5), was represented in the simulation by computing bonds between the NMC particles. A full explanation of how the bond model is applied can be found in a previous work [1]. In short, this study showed that the mechanical performance of the electrode strongly depends on the properties of the binder-additive matrix and those can be captured by the proposed bond model. The model is based on a transfer of normal and tangential forces with an additional torque transfer through bonds. The increments of normal bond force $dF_{b,n}$ and tangential bond force $dF_{b,t}$ are calculated every time step dt, along with the bond torques in normal and tangential direction, $dM_{b,n}$ and $dM_{b,t}$ [1]:

$$dF_{b,n} = -v_n \cdot S_n \cdot A \cdot dt \tag{10}$$

$$dF_{b,t} = -v_t \cdot S_t \cdot A \cdot dt \tag{11}$$

$$dM_{b,n} = -\omega_n \cdot S_n \cdot \frac{J}{2} \cdot dt \tag{12}$$

$$dM_{b,t} = -\omega_t \cdot S_t \cdot J \cdot dt \tag{13}$$

where A is the cross sectional area of the bond, J is the polar moment of inertia, v_n and v_t represent the normal and tangential linear velocities, and ω_n and ω_t correspond to the normal and tangential angular velocities. S_n and S_t are the normal and tangential area-related stiffness of the solid bond, which represent its resistance to deformation. Besides, bonds can break along the simulation if the ultimate tensile strength in normal or tangential direction is reached. Consequently, both the stiffness and the ultimate tensile strength are key for the bond model and must be accordingly defined. Since these values cannot be obtained directly via experiments, the following simulations aimed to calibrate the bond stiffness and the critical stress to define bond breakage. After the particles were inserted and the bonds were created, the top plate was set in motion and thus, the contacts between particles began to arise. The mechanical stress acting on the plate due to the direct contact with the particulate arrangement was calculated every time step. Once the mechanical stress reached the value obtained during the calendering process (P_{cal} in Table 1), the plate stopped compressing the electrode and began its displacement upwards. No effect of gravity was considered due to the fact that the particles lay in the micrometer range. The time step was 10^{-10} s, which corresponded approximately to 20% of the Raleigh time step [24]. Throughout the simulation process, several parameters were calculated (porosity, forces on the particles, number of bonds, particle positions, etc.). They provided the necessary information for the proper post-processing analysis. These outcomes will be discussed in the next section.

5. Results and discussion

5.1. Calibration and validation of the model: simulation of the calendered electrodes

Four separate simulations were performed corresponding to the produced calendered electrodes: C1, C2, C3 and C4. Fig. 4 sums up schematically the notion of the presented work. Besides, SEM images of the produced electrode and the final calendered structures (C1 to C4) are displayed along with the analogous numerically generated structures.

Fig. 5 shows the evolution of electrode porosity along the increase in mechanical stress on the plate. Each simulation was carried out five times with different initial particle arrangements for statistical reasons. For the sake of clarity, the standard deviation is not shown on the graph and only the mean values are displayed. However, the standard deviation was very low in all cases, ranging between 2 and 4.8%. As it has been mentioned, the experimental data were used to firstly calibrate the simulation in terms of bond stiffness and ultimate bond strength and secondly to validate the results. In particular, electrodes C1 and C4 were considered for the calibration, whereas C2 and C3 were regarded for the validation of the simulation procedure. Ultimately, the conditions that had to be fulfilled for structures C1 and C4 to guarantee a valid description of the binder-additive matrix were the following:

- Position B was defined by reaching the experimental normal calendering stress P_{cal} (see Fig. 3).
- The electrode porosity in this position could not be measured experimentally but it had to be lower than the final measured value; i.e. $\varepsilon_B < \varepsilon_C$.
- Likewise, the thickness in B plus half the thickness of the current collector had to be greater than half of the gap between the calendering rolls (see Table 1), i.e. $h_B + \frac{h_{Alu}}{2} > \frac{h_{Rolls}}{2}$. (It has to be noted that the reason to take the half is because the produced electrodes were double-sided coated and in the simulation only one side is considered).
- After the compaction process, the thickness (h_C) and porosity (ε_C) were determined and compared to the experimental values. The difference had to be as minimal as possible, being the requisite to achieve a deviation below 15%.

These conditions were accomplished for an area-related bond stiffness of $6 \cdot 10^{12}$ N m^{-3} and a critical stress to define the bond breakage of $2 \cdot 10^{13}$ N m^{-2}. Both magnitudes were kept the same in normal

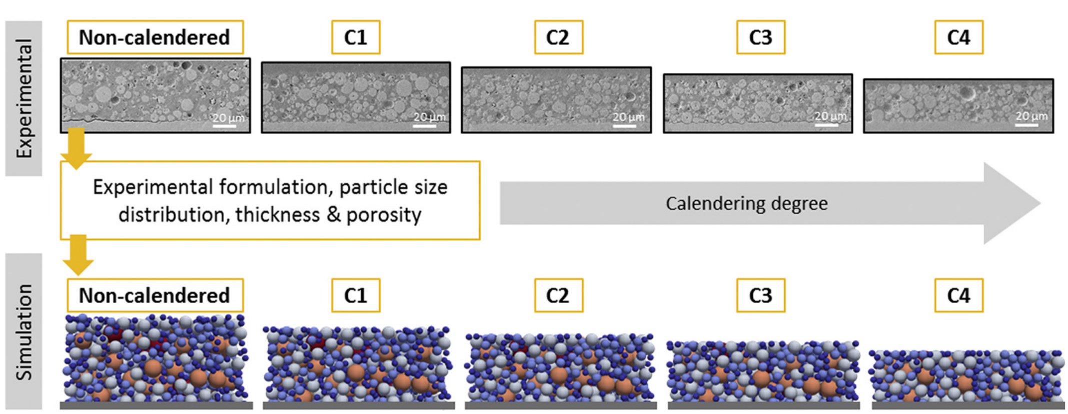

Fig. 4. Schematic of the simulation procedure. Cross-section SEM images of the final calendered electrodes and their analogous numerically generated structures (C1 to C4). The colors of the simulated assemblies are accounted for the particle radii ranging from red (coarser particles) to blue (smaller particles). Bonds between particles are not displayed.

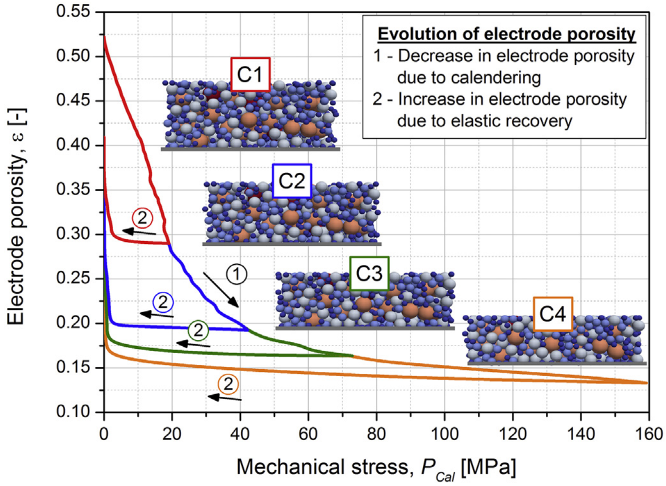

Fig. 5. Evolution of porosity along the simulation for the calendered electrodes.

and tangential direction to reduce the complexity of the calibration procedure. Once these values were calibrated for electrodes C1 and C4, simulations of electrodes C2 and C3 were carried out with the same area-related bond stiffness and ultimate bond strength. The conditions mentioned above could also be fulfilled for these structures, which proved the validation of the model. Taking for instance electrode C2, the porosity starts at 0.522 (the porosity of the non-calendered electrode) and decreases till the mechanical stress on the plate, P_{cal}, equals 42.71 MPa (see Table 1). The porosity at this point is 0.193. Afterwards, there is an elastic recovery of the electrode while the pressure is removed (see Fig. 5). The final electrode porosity is 0.368, close to the experimental value: 0.372. Overall, it can be seen that the rate of decrease in porosity at the early stages of the compaction process is faster than under high levels of compression, due to the growing contribution of particle contacts within the assembly.

Fig. 6 presents the values of ε_C obtained for each simulated electrode as well as $\varepsilon_{C,exp}$, the experimental porosity measured after the calendering process. The deviation between ε_C and $\varepsilon_{C,exp}$ is negligible for electrodes C1 and C2 and quite small for electrode C3, concluding that the method can correctly represent the electrode mechanical behavior under the respective levels of calendering stress. However, the deviation between experimental and computed values is a bit higher for electrode C4. Precisely, ε_C reached in the simulation was 0.247 whereas the experimental porosity was 0.217, which means the plasticity of this structure could not be fully captured by the simulations. It is believed that under such levels of stress, the mechanics of the electrode becomes more complex and, as a consequence, the simulation method falls short. In particular, the growth of cracks and even fracture of particles might need to be considered in order to refine the method and be able to accurately simulate high-load levels of calendering. Besides, it may be also possible that the PVDF binder reforms under high loads leading to a decrease in the elastic recovery and thus, lower values of porosity. So far, this aspect was not considered in the simulation. Nevertheless, it must be recalled that in reality such high coating densities are rarely targeted, especially in terms of high power density, due to the consequent hindrance of Li-ion transfer that would arise in the electrode [5]. The method thereby proves to be in good agreement with experimental data within a relevant range for electrode production. Additionally, Fig. 6 presents the values of ε_C obtained for each simulated electrode when performing the simulations with the Hertz contact model. It can be seen that the porosity is over-estimated and it is not possible to reach the final structural properties.

In order to compare both contact models, Fig. 7 displays the corresponding porosity curves for all simulated structures. It can be discerned that with the Hertz contact model, the corresponding levels of mechanical stress are reached more rapidly. In fact, ε_B is greater than the obtained experimental ε_C for electrodes C2, C3 and C4, which is physically incorrect. Therefore, the implementation of the elasto-plastic contact model is adequate and necessary for the analysis of NMC-based electrodes via DEM.

Fig. 8 depicts the specific elastic recovery, calculated as a function of the initial non-calendered thickness:

$$ER = \frac{h_C - h_B}{h_A} \tag{14}$$

where h_C is the final thickness, h_B is the in-between the roll thickness and $h_A$corresponds to the initial thickness. Overall, it can be seen that the elasticity decreases with increasing P_{cal}, an outcome that has also been reported on NMC electrodes by Zheng et al. [3]. Yet, electrode C1 presents a lower recovery than C2. The reason may be that in this structure, the external load might not be high enough to reach an elastic regime and a series of rearrangement mechanisms take place, such as rotation or percolation of smaller particles. Besides, the plastic

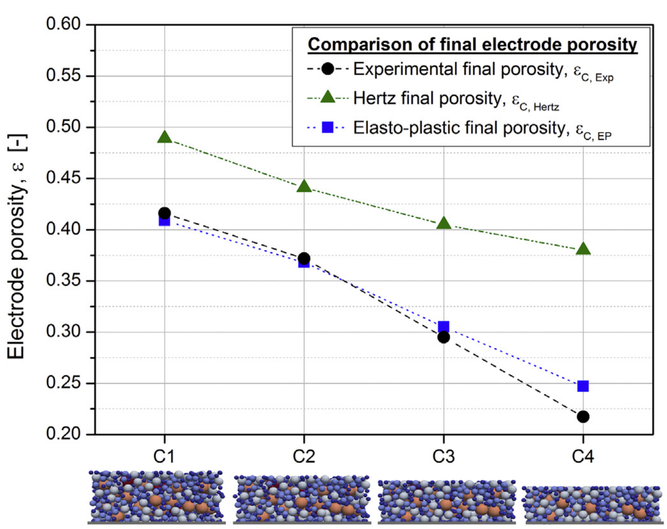

Fig. 6. Final electrode porosity obtained experimentally after the calendering process ($\varepsilon_{C,Exp}$), via simulations with the elasto-plastic contact model combined with the bond model ($\varepsilon_{C,EP}$) and via the Hertz contact model combined with the bond model ($\varepsilon_{C,Hertz}$).

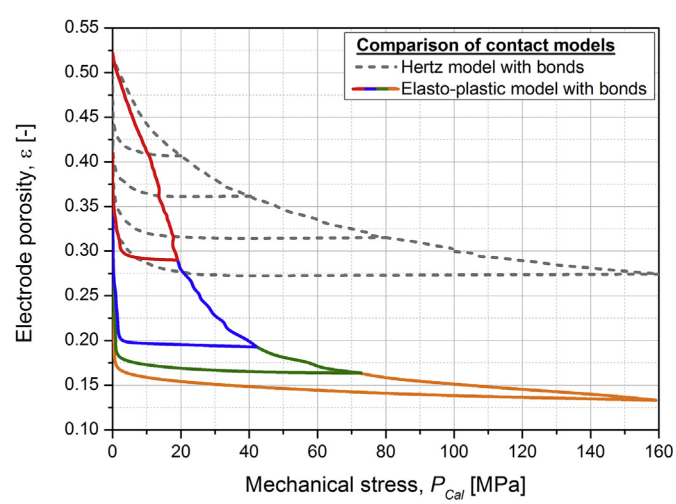

Fig. 7. Contact model comparison. Colored lines represent the proposed elasto-plastic contact model with bonds and black dotted lines correspond to the porosity course using the Hertz contact model with bonds. Same input parameters were used in both cases.

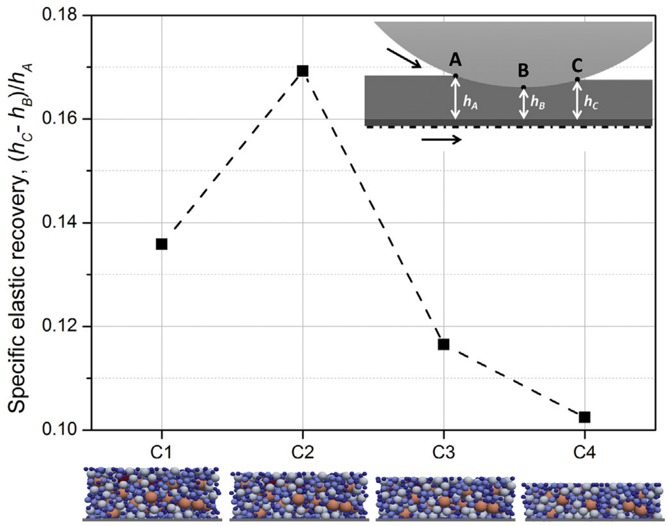

Fig. 8. Specific elastic recovery, ER, for each numerically calendered structure.

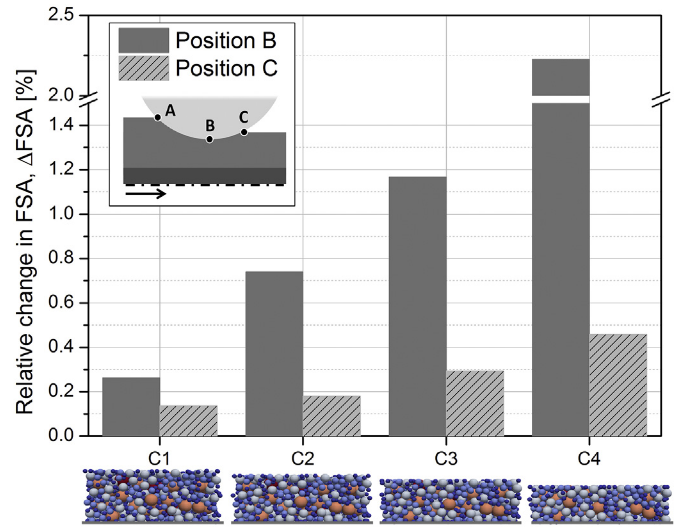

Fig. 9. Change in the specific free surface area (FSA) for the calendered electrodes at position B (calendering stress is reached) and C (final structure) related to the initial FSA.

deformation of the binder plays also a role when the overlap between particles in contact is not prominent. In electrode C2, the load level becomes sufficient to reach all particles and the elastic deformation dominates the densification process.

5.2. Structural and mechanical characterization via DEM

One of the main advantages of simulations is that they can provide practical feedback about a system and possibly predict its real behavior. This contributes to a reduction in the number of experiments, which leads to a decrease in raw materials, resources and time. Another key benefit of simulations is that they can provide deeper knowledge about a system at different levels of abstraction. By understanding the microlevel, it is easier to counteract the complexity of the macrolevel. In this section, this aspect is taken into consideration and several structural and mechanical features at particle level are shown in view of better understanding the overall electrode response during the calendering process.

5.2.1. Specific free surface area

During the operation of a lithium-ion battery, charge transfer reactions take place at the surface between active material particles and electrolyte. As already reported [7,25], a large free surface area (FSA) allows to maintain a small local overpotential at higher charging and discharging rates, which reduces activation losses. As a consequence, an increase in FSA leads to an increment in charge transfer at the expense of a decrease in electrical transfer associated with a reduction of contact area. In order to investigate the evolution of this important property along the calendering process, the specific free surface area of every particle, FSA_p, was computed as:

$$FSA_p = \frac{4\pi r_p^2 - \sum_1^{n_c} A_{c,i}}{\frac{4}{3}\pi r_p^3} \tag{15}$$

Where r_p corresponds to the radius of the particle and $\sum_1^{n_c} A_{c,i}$ is the sum of all areas related to every contact the particle is involved in. Correspondingly, the total specific FSA was determined for each simulation as the sum of the FSA_p of all particles. Fig. 9 shows the relative change in FSA; i.e. the difference between the initial and the final FSA related to the initial FSA of the non-calendered electrode:

$$\Delta FSA\ [\%] = \frac{FSA_A - FSA_x}{FSA_A}\ 100; x = B \text{ or } C \tag{16}$$

As expected, the reduction in FSA increases with higher levels of calendering (Fig. 9). Regarding the tendency of a single electrode, the final FSA in position C is greater than the FSA in position B due to the elastic recovery of the structure.

5.2.2. Coordination number and number of bonds

Even though a large specific active area is targeted in light of a higher electrochemically active area for lithium intercalation, it must not be forgotten that there are in fact two important conduction phenomena during the operation of a battery. The electrical conductivity plays a key role in cell performance as well and is directly related to the connectivity within the electrode. In this context, the coordination number (CN) and the number of bonds per particle were tracked along the simulations. It must also be underlined that this second parameter is extremely significant for the electrode assembly. As already cited, the mechanical integrity of the composite is directly linked to the

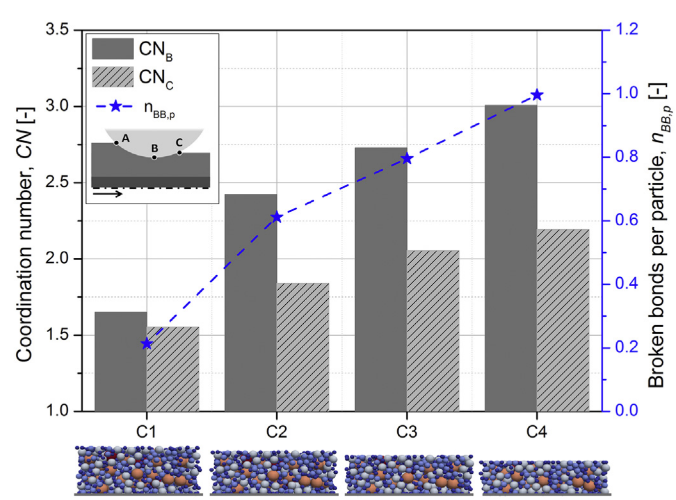

Fig. 10. Coordination number for the calendered electrodes at position B (calendering stress is reached) and C (final structure) and resulting number of broken bonds per particle.

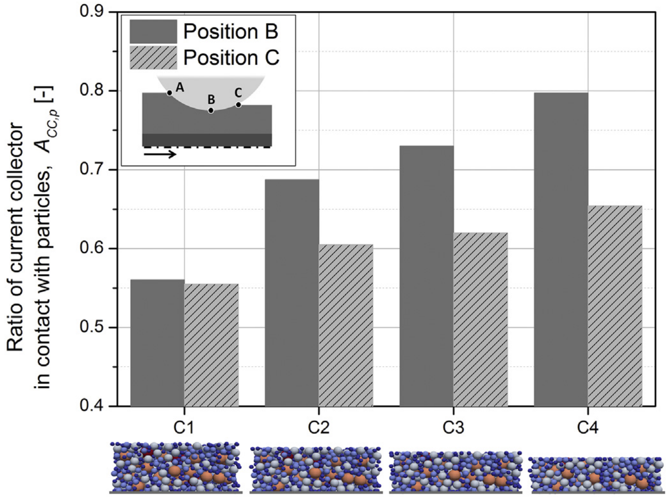

Fig. 11. Ratio of current collector surface in contact with NMC particles at position B (calendering stress is reached) and C (final structure).

mechanical performance of the binder [26]. In the presented simulation method, this is interpreted in terms of number of bonds and bond stiffness. The electrodes in this research had the same formulation since the only difference among them was the calendering degree they were subjected to. Therefore, the bond stiffness was the same for all simulations and the parameter under study was the number of broken bonds. Fig. 10 depicts the CN, which was calculated dividing the total number of particle contacts, n_c, by the number of particles, n_p:

$$CN = \frac{n_c}{n_p} \tag{17}$$

At the beginning of the simulation (position A), the particles were not in direct contact (CN = 0). As the plate begins moving downwards with the consequent decrease in porosity, contacts between particles start to arise. Hence, electrode C4 presents a higher CN than C1 ($CN_{C4} = 2.19$; $CN_{C1} = 1.55$). During the displacement of the plate from position B to C, it can be seen that in all cases the CN increases slightly owing to inertial forces and the elastic contribution of the particles and the bonds. Moreover, the average number of broken bonds per particle is also presented in Fig. 10, which was determined by the total number of broken bonds at the end of the simulation (position C), n_{BB}, divided by the number of particles:

$$n_{BB,p} = \frac{n_{BB}}{n_p} \tag{18}$$

At the beginning (position A), the average number of bonds per particle was 3.44, the maximum value along the simulation since bonds were assumed to break but not reform. As shown in Fig. 10, bonds break during the calendering process: the higher the applied calendering stress, the greater the number of broken bonds in position C. This takes places as a consequence of reaching the critical normal or tangential strength.

5.2.3. Contact area between NMC particles and the current collector

Another key aspect influencing the overall performance of a cell is the contact between the particles and the current collector. In this regard, a higher interface between the current collector and the particles results in a better electrical contact and thus, a lower electrical resistance [27]. Furthermore, electrodes have to withstand mechanical stresses during further production steps such as cutting, winding and electrolyte wetting as well as during cell operation [5]. Taking this into account, a proper contact must be ensured to avoid local delamination of the coating and consequent stress inhomogeneities, which would directly affect the battery life [28–30]. The calendering step promotes this feature because the particles are forced into the current collector which is consequently deformed. As a result, the interface between both components increases since the NMC particles can be pushed into the aluminum substrate even under moderate levels of calendering. Fig. 11 exhibits the ratio of the current collector's surface area in contact with NMC particles, $A_{CC,p}$, calculated as:

$$A_{CC,p} = \frac{\sum A_{CC,p}}{A_{CC}} \tag{19}$$

where $\sum A_{CC,p}$ is the sum of all contact areas between NMC particles and the current collector surface and A_{CC} is the total surface area of the current collector: 22500 μm^2, as depicted in Fig. 3b. A large portion of aluminum substrate is covered by particles in electrodes C3 and C4, around 62% and 65% respectively. Approximately 56% and 61% of the current collector was in contact with NMC for electrodes C1 and C2, accordingly. Therefore, it can be verified via simulations that the calendering step is favorable in this matter. Additionally, the elastic recovery of the coating has an effect on this feature and a reduction can be seen when positions B and C are compared for all assemblies.

5.2.4. Fabric tensors and stress within the electrode

Several scalar parameters such as the porosity or the particle size distribution are fundamental in characterizing the electrode structure. Nevertheless, scalar quantities are often insufficient since they do not provide information about their directionality. For instance, it has already been demonstrated that anisotropic structures may show high values of through-plane tortuosity, perpendicular to the current collector, which can negatively impact the battery performance [31]. Billaud et al. described a method in which graphite particles, functionalized with Fe_3O_4 particles, were aligned using an external magnetic field perpendicular to the current collector. Compared to a reference electrode with the same scalar porosity, the anisotropic electrode showed a three-fold improvement of its specific capacity when cycled at 1C [32]. It is therefore evident that vectorial quantities are required to fully characterize any directionally-dependent arrangement of the microstructure. In fact, the transport phenomena within the electrode are strongly influenced not only by the connectivity of the particles and the interstitial space but also by the directionality of these contacts.

Taking this into consideration, the fabric tensors of the produced electrodes were determined to uncover whether the microstructures exhibited an isotropic distribution or some degree of directional

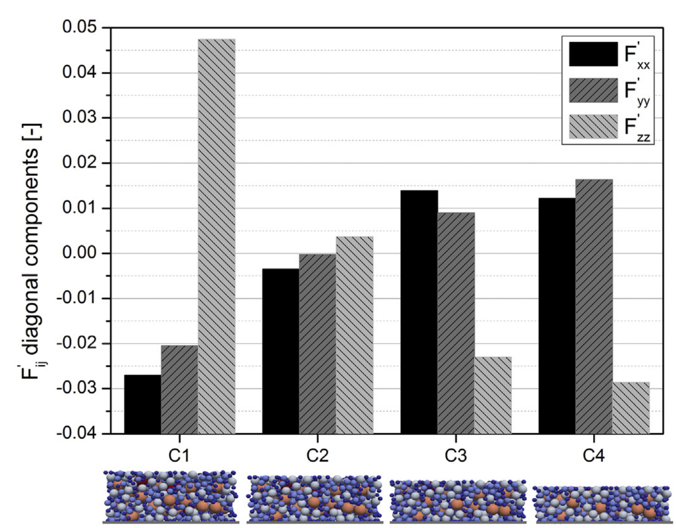

Fig. 12. Deviatoric diagonal components for the calendered electrodes at position C (final structure).

preference. Fabric tensor analysis is a numerical approximation developed by Satake [33] and Kanatani [34], which is widely used in the field of granular matter to represent contact distribution data [35–38]. The normalized distribution of contacts within the particulate assembly was described by a symmetric second-order fabric tensor F, which in index notation is given by:

$$F_{ij} = \frac{1}{N}\sum_{\alpha=1}^{N} n_i^{\alpha} n_j^{\alpha} \tag{20}$$

N represents the total number of unit contact normals $\overline{n}$ in the RVE and n_i^{α} and $n_j^{\alpha}$are the projections of the α^{th} unit vector $\overline{n}$ with respect to the x_i Cartesian coordinate. Considering all particles were spherical in the simulation, the contact normals were defined as the vector between centers of connected particles. Detailed information about this calculation can be found in [33]. Subsequently, the second-order fabric tensor was decomposed into its isotropic and deviatoric parts, $\widehat{F_{ij}}$ and F_{ij}', respectively:

$$F_{ij} = \widehat{F_{ij}} + F_{ij}' = \frac{1}{3}\delta_{ij} + F_{ij}' \tag{21}$$

Where δ_{ij} is the Kroenecker delta [33]. Whereas the isotropic part presents no directional preference of contacts, the deviatoric part can be used to quantify microstructural changes. If F_{ij}' contains only zeros, no directional preference can be identified in the distribution of contact normals. Thus, the distribution is considered statistically uniform or isotropic. In the case that one principal value of F_{ij}' is distinct while the other two are repeated, the distribution is transversely isotropic. Lastly, if the three principal values of F_{ij}' are different, the distribution is found to be orthotropically symmetric [33].

F_{ij}' of the final electrode structures (position C) were analyzed and the results are presented in Fig. 12. The F_{zz}' corresponds to the deviatoric diagonal in the normal direction of compaction whereas F_{xx}' and F_{yy}' represent the directions perpendicular to the compressing plate (see Fig. 3). No major differences can be discerned between the perpendicular diagonal magnitudes, which translates into homogeneity in the directionality of the particle contacts throughout the x-y plane. This makes sense if the periodic boundary conditions of the RVE are recalled. F_{zz}' on the other hand presents a progressive reduction if all four electrodes are compared. For electrode C1, its value is clearly higher than F_{xx}' and F_{yy}', hence, normal directionality is preferred. This is because at early stages of calendering, the movement of the plate entirely dominates the contacts within the particulate assembly. For electrode C2, the difference between F_{zz}' and the perpendicular diagonals clearly diminishes and the three components are approximately zero. There is a slight anisotropy pointing in the normal direction as the preferable one but it can be said that this electrode almost presents a state of isotropy. Further compaction (electrodes C3 and C4) implies higher levels of particle constraint and therefore, the contacts in perpendicular direction become more numerous than the ones in normal direction. As a result, F_{zz}' shifts towards negative values while F_{xx}' and F_{yy}' increase moderately leading to an inhomogeneous fabric. This fact is fundamental because as cited, the induced anisotropy affects the anisotropy of the transport properties [39].

As already stated by Mukhopadhyay et.al [26], stress development in electrode materials is one of the primary causes for capacity fade and eventual failure of Li-ion batteries. Apart from the stresses due to internal chemical processes occurring during electrochemical cycling, the external mechanical loads acting along the production steps must also be accounted for. This fact being remarked, calendering must be decidedly acknowledged. In this study, the stress evolution was implemented in terms of microscopic forces acting on individual particles. Firstly, the viral stresses acting on single NMC particles were calculated according

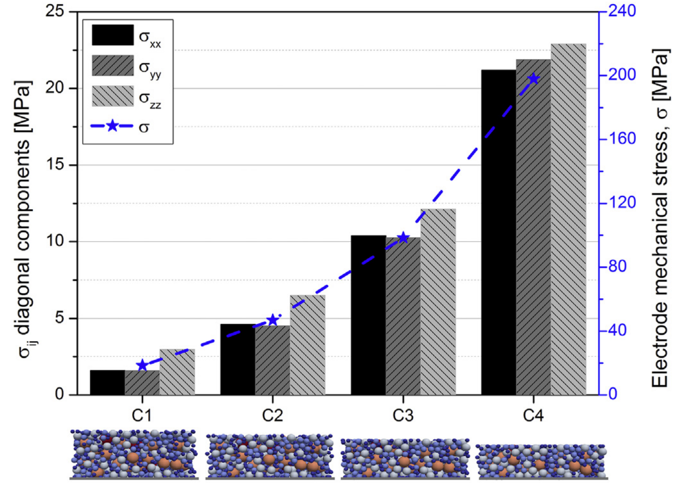

Fig. 13. Diagonal components and total magnitude of the mechanical stress within the electrode at position B (calendering stress is reached) for the calendered electrodes.

to [33] with the extension that the contribution of bond forces was likewise considered. The result was a symmetric per-particle tensor, σ_{ij}, whose components were given by:

$$\sigma_{ij} = \frac{1}{2}\sum_{1}^{N_p}\left(r_{1i}F_{1j} + r_{2i}F_{2j}\right) + \frac{1}{2}\sum_{1}^{N_b}\left(r_{1i}F_{1j} + r_{2i}F_{2j}\right) \tag{22}$$

where N_p and N_b correspond to the number of contacts and bonds, respectively; i and j take values of x,y and z to evaluate all components of the tensor; r_1 and r_2 are the positions of the two particles in the pairwise or bond interaction; and F_1 and F_2 are the forces acting on them due to the corresponding interaction. As it is implied in Eq. (22), the tensor has units of stress multiplied by volume. Thus, the diagonals of each per-particle stress tensor were summed and the total macroscopic stress was averaged over the total volume of the simulated RVE. Fig. 13 shows the diagonals as well as the total stress within the electrode when P_{cal} is reached (position B). It was decided to analyze this position because it is the critical point in terms of mechanical constraint. As predicted, the higher the calendering degree, the greater the stress generated within the electrode. It is remarkable that C1 developed a σ_{zz} which was twice as high as the diagonals in x and y direction. The difference between the normal and the perpendicular stress diagonals clearly diminishes, being σ_{zz}/σ_{xx} approximately 1.4 for C2, 1.17 for C3 and 1.08 for C4. This tendency is by all means related to the directionality of the contacts since the stress tensor is indeed derived from the fabric tensor. However, it would be reasonable to state that the directionality of the bonds is not favorable in normal direction on account of the fact that the whole real thickness of the electrode is simulated and the boundary conditions in x and y directions are also applied to the bonds. The bond interactions are also involved in the calculation of the macroscopic stress and therefore, the same tendency observed in Fig. 12 is not fully captured. Future research may use this information to assess the critical stress for particle breakage and establish a relation between the calendering degree and the rise of particle deterioration.

6. Conclusion

This work developed a DEM simulation model to describe the behavior of lithium-ion electrodes during the calendering production step from the micro- to the macroscale. In fact, NMC active material particles were experimentally characterized in order to establish a proper DEM contact model. The obtained nanoindentation curves deviated increasingly from the theoretical Hertz curve with incrementing the compression force. This aspect made it clear that under high levels of compression, this contact model would overestimate the real behavior of NMC particles. Therefore, an elasto-plastic model based on the work of Thornton and Ning [16] was adopted and the yield ratio of a host of 40 particles was determined and used as the outset of the model. Moreover, the contact model was combined with a bond model [1] to represent the additive-binder matrix, which plays a fundamental role in the macromechanical properties of the electrode. Input parameters were gathered from in-house experiments to rely on the accuracy of the simulations. For the purpose of calibrating the model, one NMC-based cathode was produced and subjected to two different calendering levels (C1 with coating density of 2.61 g cm^{-3} and C4 with a coating density of 3.49 g cm^{-3}, see Table 1). By presetting several conditions, the numerical simulations were calibrated for a bond stiffness of $6 \cdot 10^{12}$ N m^{-3} and a ultimate bond strength of $2 \cdot 10^{13}$ N m^{-2}. Secondly, the simulation procedure was validated for two other electrodes (C2 with coating density of 2.80 g cm^{-3} and C3 with a coating density of 3.15 g cm^{-3}, see Table 1). In light of the results, it could be demonstrated that the model is adequate to describe the macroscopic behavior of the electrode along the calendering step. Particularly interesting from the simulation method is the possibility to reproduce the elastic recovery of the electrode after releasing the calendering stress. This feature cannot be obtained experimentally and is fundamental to better control the calendering process. It was shown that the elastic recovery of the electrodes, which was proved to depend on the calendering degree, ranged from 10.25% up to almost 17%.

Furthermore, the numerical investigations contributed to get deeper knowledge about the calendering process. In the scope of this contribution, several structural and mechanical features were examined in view of better understanding the overall response of a NMC-based electrode during calendering. This detailed research quantitatively studied the evolution of the change in specific free surface area and number of bonds as well as the increment in coordination number and contact area between active material and current collector with increasing calendering load. Likewise, it was proved that the contacts in normal direction dominated the structure of electrode C1 whereas electrode C2 presented an almost state of isotropy. Further compaction led to an inhomogeneous fabric (electrodes C3 and C4). The most critical position, when the calendering load is reached, was analyzed in terms of mechanical stress generated within the structure. An increase of this feature could also be quantified for the corresponding structures. This work constitutes a first part of what is aim to be a numerical overview of the interactions between the electrode structure and its properties during the calendering process. The second part, presented in a following publication, focuses on defining these relations by taking advantage of the simulations. These properties are evaluated along the first electrochemical cycles, which also contributes to point to more suitable electrode structures. Furthermore, there are also potentially new features that may be implemented in the near future such as the possibility to regard particle breakage or the feasibility to reform binder bonds - an interesting aspect to be investigated both experimentally and numerically. On the whole, this simulation method offers the possibility to gain insight into the overall electrode behavior during the calendering process in view of predicting key cell properties and designing improved materials.

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